

AI Clinic

AI for Healthcare Compliance and Regulations

Interactive Dashboard and Comparative Regulatory Analysis

Project Report

Supervisor: Dr. Anuradha Kar

March 2026

A structured synthesis of AI healthcare regulation across 20 countries, seven compliance themes, and an interactive Streamlit analytics dashboard. Includes literature review, dataset design, implementation, and findings.

Abstract

Artificial intelligence (AI) is reshaping healthcare through diagnostic imaging, digital pathology, genomic interpretation, drug discovery support, and clinical decision support embedded in hospital workflows. These capabilities create substantial regulatory complexity: software may qualify as Software as a Medical Device (SaMD), process sensitive health data under divergent privacy regimes, and require lifecycle governance when algorithms adapt over time.

This project delivers an end-to-end research and engineering artifact comprising (1) a structured literature review of AI healthcare compliance across 20 countries in six regions; (2) a curated JSON dataset scoring seven regulatory themes on a 1–10 scale; and (3) an interactive Streamlit dashboard with geospatial visualization, comparative analytics, global trend tracking, AI use-case readiness views, and an integrated literature browser. The dashboard enables policymakers, researchers, and developers to compare jurisdictional maturity at a glance and drill into country-specific regulatory narratives.

Key findings include convergence toward risk-based SaMD classification and comprehensive data protection modeled on GDPR, persistent disparities in enforcement capacity between advanced and emerging economies, and growing emphasis on post-market surveillance and liability frameworks for adaptive AI. The EU AI Act represents a watershed in binding AI governance. Clinical adoption trends in radiology, pathology, genomics, and public health surveillance amplify the need for transparent validation, representative datasets, and Total Product Lifecycle oversight.

Keywords: AI healthcare regulation, SaMD, compliance dashboard, data privacy, clinical validation, EU AI Act, post-market surveillance, Streamlit.

Contents

Abstract	i
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Project Objectives	1
1.4 Scope and Limitations	2
1.5 Report Organization	2
2 Literature Review	3
2.1 Regulatory Frameworks for AI Medical Devices	3
2.1.1 Software as a Medical Device	3
2.1.2 Risk-Based Classification	3
2.1.3 Adaptive Algorithms and Lifecycle Regulation	3
2.2 Data Privacy and Governance	4
2.2.1 The GDPR Paradigm	4
2.2.2 Post-GDPR Global Wave	4
2.2.3 Health-Specific Provisions	4
2.3 Clinical Validation, Safety, and Transparency	4
2.4 Ethics and Accountability	4
2.5 Post-Market Surveillance and Liability	5
2.6 Comparative Regional Synthesis	5
3 Research Methodology	6
3.1 Literature Search Strategy	6
3.2 Dataset Construction	6
3.3 Dashboard Development Methodology	6
3.4 Deliverable Pipeline	7
4 Dataset Design	8
4.1 Schema Overview	8
4.2 Seven Compliance Themes	8
4.3 Countries and Regions	8
4.4 Global Trends	8
4.5 Complete Score Matrix	9

5	Interactive Dashboard Architecture	11
5.1	Technology Stack	11
5.2	Application Structure	11
5.3	User Interface Components	11
5.3.1	Sidebar Filters	11
5.3.2	Executive Snapshot	11
5.3.3	Seven Analytical Tabs	12
5.4	Use-Case Readiness Model	12
5.5	Deployment	12
6	Results and Comparative Analysis	13
6.1	Global Theme Averages	13
6.2	Regional Patterns	13
6.3	Maturity Labels	13
6.4	Device Approval Counts	13
6.5	Key Regulatory Milestones	13
7	AI Use Cases in Clinical, Hospital, and Public Health Workflows	15
7.1	Radiology and Imaging Diagnostics	15
7.2	Pathology and Digital Pathology	15
7.3	Genomic Analysis and Precision Medicine	15
7.4	Drug Discovery and Clinical Development Support	15
7.5	Hospital and Clinic Clinical Decision Support	16
7.6	Public Health Surveillance	16
7.7	Foundation and Generative AI	16
8	Global Trends and Future Directions	17
8.1	Convergence Trends	17
8.2	Emerging Challenges	17
8.3	Recommendations	17
9	Conclusion	18
	Data Sources and Primary References	19
	References	20

List of Tables

2.1	Risk-based medical device classification systems (selected jurisdictions)	3
4.1	Compliance theme definitions	8
4.2	Country compliance scores (1–10) across seven themes	9
4.3	Global trends in AI healthcare regulation	9
5.1	Implementation stack	11

Chapter 1

Introduction

1.1 Background

The integration of AI into healthcare represents one of the most consequential technological shifts of the 21st century. AI-based SaMD assists clinicians in radiology, pathology, ophthalmology, cardiology, and genomics, among other specialties [1], [2]. Major regulators have authorized hundreds of AI/ML-enabled devices, reflecting exponential growth in the field [3], [4].

Unlike traditional medical devices, AI systems can be adaptive, opaque, and dependent on training data that may embed biases—raising novel regulatory questions that existing frameworks were not designed to address [5], [6]. Ensuring patient safety, data privacy, algorithmic fairness, and accountability therefore requires jurisdiction-specific compliance strategies and continuous monitoring after deployment.

1.2 Problem Statement

Regulatory diversity across countries creates challenges for:

- **Developers** seeking international market access and harmonized evidence packages;
- **Researchers** studying comparative AI governance;
- **Policymakers** balancing innovation incentives with safety protections;
- **Clinicians and hospital administrators** evaluating whether AI tools meet local legal and ethical requirements.

There is a need for structured, comparative analysis presented through both scholarly synthesis and accessible interactive visualization.

1.3 Project Objectives

This project aims to:

1. Synthesize peer-reviewed and policy literature on AI healthcare regulation across 20 countries;
2. Define seven thematic dimensions of compliance and score jurisdictional maturity;
3. Build a curated, machine-readable dataset with regional metadata and global trends;
4. Implement a professional interactive dashboard for exploration and export;
5. Document methodology, architecture, and findings in this report.

1.4 Scope and Limitations

The analysis covers 20 countries across North America, Europe, East Asia, South/Southeast Asia, the Middle East, and Africa, plus Australia. Theme scores are expert-curated syntheses informed by regulatory documents and literature—they are comparative indicators, not official government ratings. AI use-case readiness scores in the dashboard are *derived* from theme weights rather than separate regulatory datasets. The report should be read as an academic and decision-support tool, not legal advice.

1.5 Report Organization

Chapter 2 presents the literature review. Chapter 3 describes research and scoring methodology. Chapter 4 documents the dataset. Chapter 5 explains dashboard design. Chapters 6–7 present results, regional analysis, and clinical use cases. Chapter 9 concludes with recommendations.

Chapter 2

Literature Review

2.1 Regulatory Frameworks for AI Medical Devices

2.1.1 Software as a Medical Device

The IMDRF framework (2014) provides foundational categorization for software that meets the definition of a medical device without being part of hardware [7]. Major jurisdictions have adopted or adapted it:

- **USA:** FDA regulates AI/ML SaMD via 510(k), De Novo, and PMA pathways, supplemented by the AI/ML SaMD Action Plan [3];
- **EU:** MDR classifies SaMD by intended purpose and risk; the EU AI Act adds high-risk AI obligations [8];
- **Japan:** PMDA administers the DASH framework [9];
- **China:** NMPA issued guiding principles for AI medical device registration [10].

2.1.2 Risk-Based Classification

A near-universal trend is proportional regulation by potential harm. Higher-risk applications (e.g., autonomous diagnosis) face stricter pre-market evaluation and post-market obligations [11]. Table 2.1 summarizes representative classification systems.

Table 2.1: Risk-based medical device classification systems (selected jurisdictions)

Region	System	Classes
USA	FDA risk-based	I, II, III
EU	MDR risk-based	I, IIa, IIb, III
China	NMPA three-tier	I, II, III
Canada	Health Canada	I, II, III, IV
Japan	PMDA/PMD Act	I, II, III, IV
Australia	TGA	I, IIa, IIb, III

2.1.3 Adaptive Algorithms and Lifecycle Regulation

AI/ML algorithms can evolve with new data. Regulators respond with lifecycle approaches rather than point-in-time approval [12], [13]. The FDA Predetermined Change Control

Plan (PCCP) allows pre-specified modifications without new submissions for each change [14]. The EU AI Act requires ongoing conformity and risk management. Good Machine Learning Practice (GMLP) principles from FDA, Health Canada, and MHRA guide development practices [15].

2.2 Data Privacy and Governance

2.2.1 The GDPR Paradigm

GDPR (2018) established global benchmarks: lawfulness, purpose limitation, data minimization, accountability, data subject rights, and Data Protection Impact Assessments. Article 22’s provisions on automated decision-making have significant implications for clinical AI [16].

2.2.2 Post-GDPR Global Wave

Comprehensive data protection laws have spread to Brazil (LGPD), India (DPDP), China (PIPL), South Africa (POPIA), Nigeria (NDPA), Kenya, Thailand, Gulf states, and Switzerland—with varying GDPR alignment.

2.2.3 Health-Specific Provisions

The USA relies on HIPAA for Protected Health Information [17]. The EU proposes a European Health Data Space for primary and secondary use. Cross-border genomic and AI validation increasingly depend on lawful transfer mechanisms and representativeness of training data [5].

2.3 Clinical Validation, Safety, and Transparency

Clinical evidence requirements for AI SaMD remain contested. Studies of FDA-cleared devices highlight limitations in external validation and subgroup performance [4]. Algorithmic bias in population health tools demonstrates real-world harm potential [18], [19].

Transparency debates pit explainable AI against interpretable-by-design models [20], [21]. Regulators increasingly expect human-readable documentation, performance monitoring, and clinician-facing intelligibility even when full algorithmic transparency is infeasible.

2.4 Ethics and Accountability

WHO ethics guidance emphasizes human oversight, transparency, accountability, inclusiveness, and sustainability [22]. Jobin et al. mapped hundreds of AI ethics guidelines globally [23]. A persistent “ethics gap” exists between voluntary principles and enforceable practice [24], [25].

Regional perspectives differ: EU rights-based approaches, US innovation-oriented voluntary frameworks, China’s state-guided development, African Ubuntu-influenced discourse, and Middle Eastern intersections of Islamic ethics with modernization.

2.5 Post-Market Surveillance and Liability

Post-market systems include FDA MedWatch, EU MDR serious incident reporting, and UK Yellow Card. AI-specific harms (silent degradation, systematic bias) may not fit traditional adverse event categories [26]. Liability frameworks largely rely on product liability and malpractice law, which may be inadequate for opaque AI [27]. The proposed EU AI Liability Directive would shift causation burdens in some cases.

2.6 Comparative Regional Synthesis

North America combines advanced agencies (FDA, Health Canada) with fragmented US federal AI law. **Europe** leads with the EU AI Act, MDR, and GDPR integration. **East Asia** shows rapid approval activity and algorithm-specific rules (China, South Korea). **South/Southeast Asia** spans Singapore's governance leadership and India's emerging digital health infrastructure. **Middle East** pairs Gulf investment with Israel's innovation ecosystem. **Africa** faces capacity constraints but benefits from mobile health innovation and new data protection laws.

Chapter 3

Research Methodology

3.1 Literature Search Strategy

A structured review used PubMed, Scopus, IEEE Xplore, Google Scholar, and SSRN, combined with regulatory sources (FDA, EMA, MHRA, NMPA, WHO, OECD) and national AI strategies. Search terms included “AI healthcare regulation,” “SaMD compliance,” “health data governance,” and “algorithmic transparency medicine.” Inclusion criteria: publications from 2018–2026 addressing AI regulation, compliance, or ethics in healthcare. Thirty-two peer-reviewed and policy references inform the synthesis.

3.2 Dataset Construction

For each of 20 countries, the project records:

- Geographic region and regulatory maturity label;
- Seven theme scores (1–10): data privacy, clinical validation, approval process, transparency, ethics, post-market surveillance, liability;
- Narrative regulatory summary and key legislation;
- Count of AI/ML medical devices approved (where reported);
- Pointers to primary regulatory sources.

Scores were assigned by thematic coding of policy documents and literature, then calibrated for cross-country comparability. Higher scores indicate more mature, AI-specific, or enforceable frameworks—not necessarily “better” innovation climates.

3.3 Dashboard Development Methodology

The dashboard follows an iterative design process:

1. Define user personas (researcher, policymaker, student presenter);
2. Map each persona to visualization needs (map, bar charts, radar, trends);
3. Implement in Streamlit with Plotly for interactivity;
4. Add executive snapshot metrics and CSV export for reproducibility;
5. Extend with AI use-case views derived from weighted theme combinations.

3.4 Deliverable Pipeline

Supporting scripts generate PDF reports (literature, full report, dashboard snapshot, data source links) and PowerPoint slides. A master script (`run_all.py`) orchestrates generation and optional dashboard launch.

Chapter 4

Dataset Design

4.1 Schema Overview

The canonical dataset is stored as `data/compliance_dataset.json`. Top-level keys include `countries`, `global_trends`, and `key_references`. Each country entry contains identifiers, regional grouping, maturity level, theme scores, device counts, and descriptive regulatory text.

4.2 Seven Compliance Themes

Table 4.1: Compliance theme definitions

Theme	Description
Data Privacy	Health data protection laws, cross-border transfer rules, DPIAs
Clinical Validation	Evidence standards, clinical evaluation, bias assessment
Approval Process	SaMD pathways, AI-specific guidance, regulatory sandboxes
Transparency	Explainability expectations, documentation, clinician intelligibility
Ethics	National AI ethics frameworks, fairness, human oversight
Post-Market	Adverse event reporting, lifecycle monitoring, update governance
Liability	Product liability, malpractice interfaces, AI-specific proposals

4.3 Countries and Regions

Twenty countries span six regions: USA and Canada; EU aggregate plus UK, Germany, Switzerland; China, India, Japan, South Korea, Singapore, Thailand; Saudi Arabia, UAE, Israel; South Africa, Nigeria, Kenya; and Australia.

4.4 Global Trends

The dataset includes structured trend objects with adoption level, emergence year, and narrative description. Trends cover radiology/pathology adoption, genomic AI, drug

discovery, hospital CDS deployment, and public health surveillance—linking regulatory themes to real-world product categories.

4.5 Complete Score Matrix

Table 4.2 lists all country scores. Privacy = Data Privacy; Clin. = Clinical Validation; Appr. = Approval Process; Trans. = Transparency; Eth. = Ethics; Post = Post-Market; Liab. = Liability; Devices = reported AI device approvals.

Table 4.2: Country compliance scores (1–10) across seven themes

Country	Region	Maturity	Priv.	Clin.	Appr.	Trans.	Eth.	Post	Liab.
European Union	Europe	Advanced	10	9	9	9	10	9	8
Germany	Europe	Advanced	10	9	9	8	9	9	7
United States	North America	Advanced	8	9	9	6	6	8	7
United Kingdom	Europe	Advanced	8	8	7	7	7	7	6
Canada	North America	Advanced	7	8	8	7	7	7	6
Japan	Asia	Advanced	7	8	8	6	7	8	5
Singapore	Asia	Advanced	7	7	7	8	8	6	5
Switzerland	Europe	Advanced	8	8	8	6	6	7	5
China	Asia	Advanced	7	7	7	7	6	7	6
Australia	Oceania	Moderate	7	7	7	6	7	7	5
South Korea	Asia	Advanced	7	7	8	6	6	7	5
Israel	Middle East	Advanced	6	8	7	6	6	6	5
Brazil	South America	Developing	7	5	5	5	5	5	5
UAE	Middle East	Emerging	5	5	5	5	5	4	3
Saudi Arabia	Middle East	Emerging	5	4	5	5	5	4	3
India	Asia	Developing	5	4	4	4	5	4	4
Thailand	Asia	Developing	5	4	4	4	4	3	3
South Africa	Africa	Emerging	6	3	3	3	4	3	3
Kenya	Africa	Early	5	2	3	2	3	2	2
Nigeria	Africa	Early	4	2	3	2	3	2	2

Table 4.3: Global trends in AI healthcare regulation

Trend	Description	Adoption	Since
Convergence on Risk-Based Regulation	Most jurisdictions adopting risk-based classification of AI medical devices, influenced by IMDRF framework.	High	2017
Comprehensive Data Protection Laws	Post-GDPR wave of national data protection laws with health data as sensitive category.	High	2018
AI-Specific Legislation	Growing trend toward dedicated AI laws (EU AI Act leading, followed by Brazil, Canada, South Korea proposals).	Medium	2021

Trend	Description	Adoption	Since
International Harmonization Efforts	IMDRF, GMLP (FDA-Health Canada-MHRA), and WHO guidance driving convergence.	Medium	2019
Ethical AI Frameworks	Proliferation of national AI ethics principles, mostly voluntary, increasingly being legislated.	High	2018
Regulatory Sandboxes for Health AI	Controlled testing environments for innovative AI devices before full regulatory compliance.	Low	2020
Real-World Evidence Acceptance	Growing regulatory acceptance of real-world data and evidence for AI device evaluation.	Medium	2019
Adaptive Algorithm Governance	Frameworks for managing continuously learning AI systems (e.g., FDA Predetermined Change Control Plan).	Low	2021
AI Adoption in Radiology and Digital Pathology Workflows	AI image analysis is increasingly deployed for triage and decision support in radiology and pathology, driving demand for clinical validation, transparency for clinicians, and post-deployment monitoring.	High	2021
Genomic AI for Precision Medicine (Data Governance Focus)	Genomic interpretation and clinical decision support are expanding, with governance emphasizing privacy, representative datasets, and controlled cross-border data use.	Medium	2020
AI-Augmented Drug Discovery and Clinical Development Support	AI supports discovery and development workflows and increasingly influences clinical evidence generation, increasing focus on documentation, auditability, and liability assumptions.	Medium	2021
Hospital Deployment of Clinical Decision Support (Workflow Integration)	Hospitals and clinics integrate AI into electronic workflows, making Total Product Life-cycle governance and user training central to safe deployment.	High	2022
Public Health Surveillance and Outbreak Support with AI	AI systems are being explored for population-level surveillance and outbreak support, requiring accountability mechanisms because impacts extend beyond individual patients.	Medium	2020

Chapter 5

Interactive Dashboard Architecture

5.1 Technology Stack

Table 5.1: Implementation stack

Component	Technology
Runtime	Python 3.9+
UI framework	Streamlit
Visualization	Plotly (Express and Graph Objects)
Data processing	Pandas
Data store	JSON (version-controlled)
Static exports	fpdf2, python-pptx

5.2 Application Structure

The main application (`app.py`) loads the JSON dataset at startup, normalizes country records into a Pandas DataFrame, and renders a wide-layout interface with custom CSS (gradient background, metric cards, professional typography).

5.3 User Interface Components

5.3.1 Sidebar Filters

Users filter by region, maturity level, and minimum theme score. A download button exports the filtered subset as CSV for offline analysis.

5.3.2 Executive Snapshot

Above the tabbed content, summary cards highlight the highest- and lowest-scoring countries, the largest inter-theme gap, and a bar chart of average theme scores globally.

5.3.3 Seven Analytical Tabs

1. **World Map** — Choropleth of composite compliance scores;
2. **Country Comparison** — Side-by-side bar and radar charts;
3. **Theme Analysis** — Heatmaps and distribution plots;
4. **Global Trends** — Trend cards and adoption timeline;
5. **Country Details** — Deep-dive narratives per jurisdiction;
6. **AI Use Cases & Trends** — Derived readiness for radiology, pathology, genomics, drug discovery;
7. **Literature Review** — Keyword frequency, searchable references, full markdown render.

5.4 Use-Case Readiness Model

The AI Use Cases tab computes derived scores by weighting compliance themes relevant to each clinical domain. For example, radiology readiness weights clinical validation and post-market surveillance heavily; genomics emphasizes data privacy; drug discovery emphasizes approval process and transparency. This approach links regulatory maturity to deployment contexts while making derivation explicit to users.

5.5 Deployment

Local deployment:

```
python -m streamlit run app.py
```

Default URL: <http://localhost:8501>. The dashboard requires no external API keys; all data is bundled locally.

Chapter 6

Results and Comparative Analysis

6.1 Global Theme Averages

Across all 20 countries, approval process and data privacy themes tend to score highest on average, reflecting widespread adoption of medical device frameworks and post-GDPR privacy laws. Post-market surveillance and AI-specific liability often lag, indicating lifecycle governance as the emerging frontier.

6.2 Regional Patterns

Europe and North America consistently rank among top performers on composite scores, driven by mature agencies and comprehensive legislation. **East Asia** shows strong approval throughput (especially South Korea and China) with variable privacy maturity. **Middle East** jurisdictions demonstrate rapid framework development aligned with national AI strategies. **African** countries score lower on AI-specific regulation but show progress via data protection laws and mobile health innovation.

6.3 Maturity Labels

Countries are labeled Advanced, Developing, or Emerging based on composite scores and qualitative regulatory capacity. Advanced jurisdictions typically have AI-specific SaMD guidance, active device approvals, and enforceable post-market systems. Emerging jurisdictions may have general medical device law without AI-tailored pathways.

6.4 Device Approval Counts

Reported AI device counts correlate with regulatory agency capacity and market size but do not alone indicate governance quality. The dashboard presents counts alongside theme scores to avoid single-metric rankings.

6.5 Key Regulatory Milestones

- EU AI Act (2024) — binding high-risk AI requirements;

- FDA AI/ML SaMD Action Plan (2021) and PCCP guidance (2023);
- WHO AI ethics guidance (2021);
- GMLP harmonization (2021);
- National data protection waves (PIPL, DPDP, POPIA, NDPA).

Chapter 7

AI Use Cases in Clinical, Hospital, and Public Health Workflows

Regulatory analysis must connect to deployment contexts. Current literature and industry practice highlight seven major use-case clusters.

7.1 Radiology and Imaging Diagnostics

AI supports triage, detection, and decision support in imaging workflows. Compliance emphasis centers on multi-site clinical validation, subgroup performance, transparency for radiologists, and continuous monitoring after PACS integration [2], [4].

7.2 Pathology and Digital Pathology

Whole-slide imaging plus AI accelerates diagnosis but depends on labeling quality and scanner standardization. Governance themes include sensitive tissue data handling, evidence quality, and lifecycle surveillance when models retrain on institutional data.

7.3 Genomic Analysis and Precision Medicine

Variant interpretation and risk stratification require stringent privacy and cross-border data governance. Representativeness of genomic reference panels affects fairness; regulators scrutinize intended use claims tied to ancestry and clinical context.

7.4 Drug Discovery and Clinical Development Support

AI influences target identification, candidate ranking, and trial design. Regulatory attention grows around reproducibility, audit trails, and how computational evidence supports safety documentation—bridging pharma quality systems and SaMD logic.

7.5 Hospital and Clinic Clinical Decision Support

Health systems embed AI CDS in EHR workflows. The shift from one-time approval to Total Product Lifecycle thinking is critical: software updates, population drift, and workflow integration can alter real-world performance [13], [26].

7.6 Public Health Surveillance

AI for outbreak detection, contact tracing analytics, and population risk scoring extends governance beyond individual patients. Accountability, proportionality, and human rights safeguards are essential when decisions affect communities at scale [22].

7.7 Foundation and Generative AI

Large language models and foundation models challenge classic SaMD assumptions: broad intended uses, rapid updates, and emergent behaviors require new evaluation paradigms, usage policies, and monitoring—areas where global regulation is still forming.

Chapter 8

Global Trends and Future Directions

8.1 Convergence Trends

1. Risk-based SaMD classification is near-universal among mature regulators;
2. Comprehensive data protection legislation continues to spread;
3. International harmonization via IMDRF, WHO, and GMLP accelerates;
4. Ethical frameworks shift from voluntary principles to binding law [8].

8.2 Emerging Challenges

Foundation models, continuous learning, Global South capacity gaps, and data localization threaten fragmented AI development. OECD and NIST frameworks offer voluntary risk management tools that may inform future binding rules [28], [29].

8.3 Recommendations

1. Expand IMDRF/WHO coordination for AI-specific clinical evidence methods;
2. Build regulatory capacity in developing countries via technical assistance;
3. Operationalize ethics through measurable fairness and monitoring standards;
4. Proactively update liability frameworks before harm cases proliferate;
5. Maintain public dashboards and open datasets for transparent comparative governance research.

Chapter 9

Conclusion

The global regulatory landscape for AI in healthcare is characterized by rapid evolution, converging principles, and persistent diversity. While risk-based classification, data protection, and ethical governance have emerged as near-universal themes, significant disparities remain in enforcement capacity and AI-specific legislation.

This project contributes a reproducible pipeline from literature synthesis to interactive analytics. The Streamlit dashboard translates complex jurisdictional differences into explorable visual form suitable for academic presentation and stakeholder briefing. Future work may incorporate live regulatory feeds, formal expert Delphi scoring, and expanded country coverage.

Success in governing AI healthcare will depend on balancing innovation with safety, strengthening international harmonization, and ensuring frameworks adapt to foundation models, continuous learning, and equitable access across regions.

Data Sources and Primary References

Primary regulatory and policy sources used in dataset curation include:

- FDA AI/ML SaMD Action Plan and PCCP guidance;
- EU AI Act, MDR, and GDPR texts;
- WHO Ethics and Governance of AI for Health;
- OECD.AI Policy Observatory;
- IMDRF SaMD framework;
- GMLP guiding principles;
- NIST AI Risk Management Framework;
- EMA, MHRA, PMDA, NMPA, and national AI strategy documents.

A standalone PDF listing URLs is generated by `generate_links_pdf.py` as `Data_Source_Links.pdf`.

References

- [1] E. J. Topol, “High-performance medicine: The convergence of human and artificial intelligence,” *Nature Medicine*, vol. 25, pp. 44–56, 2019.
- [2] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, “Ai in health and medicine,” *Nature Medicine*, vol. 28, pp. 31–38, 2022.
- [3] U.S. Food and Drug Administration, *Artificial intelligence/machine learning (ai/ml)-based software as a medical device (samd) action plan*, <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device>, 2021.
- [4] E. Wu, K. Wu, R. Daneshjou, et al., “How medical AI devices are evaluated: Limitations and recommendations from an analysis of FDA approvals,” *Nature Medicine*, vol. 27, pp. 582–584, 2021.
- [5] S. Gerke, T. Minssen, and I. G. Cohen, “Ethical and legal challenges of artificial intelligence-driven healthcare,” *Artificial Intelligence in Healthcare*, pp. 295–336, 2020.
- [6] D. S. Char, N. H. Shah, and D. Magnus, “Implementing machine learning in health care — addressing ethical challenges,” *New England Journal of Medicine*, vol. 378, no. 11, pp. 981–983, 2018.
- [7] International Medical Device Regulators Forum, *Software as a medical device: Possible framework for risk categorization and corresponding considerations*, 2014.
- [8] European Parliament and Council, *Regulation (eu) 2024/1689 — artificial intelligence act*, 2024.
- [9] Pharmaceuticals and Medical Devices Agency (Japan), *Dash for samd: Regulatory framework*, 2020.
- [10] National Medical Products Administration (China), *Guiding principles for registration review of AI medical device software*, 2021.
- [11] U. J. Muehlematter, P. Daniore, and K. N. Vokinger, “Approval of artificial intelligence and machine learning-based medical devices in the USA and Europe (2015–20),” *The Lancet Digital Health*, vol. 3, no. 3, e195–e203, 2021.
- [12] B. Babić, S. Gerke, T. Evgeniou, and I. G. Cohen, “Algorithms on regulatory lockdown in medicine,” *Science*, vol. 366, no. 6470, pp. 1202–1204, 2019.
- [13] T. J. Hwang, A. S. Kesselheim, and K. N. Vokinger, “Lifecycle regulation of AI- and ML-based software devices in medicine,” *JAMA*, vol. 322, no. 23, pp. 2285–2286, 2019.
- [14] U.S. Food and Drug Administration, *Marketing submission recommendations for a predetermined change control plan for ai/ml-enabled device software functions*, 2023.

- [15] FDA, Health Canada, and MHRA, *Good machine learning practice for medical device development: Guiding principles*, 2021.
- [16] B. Goodman and S. Flaxman, “European union regulations on algorithmic decision-making and a “right to explanation”,” *AI Magazine*, vol. 38, no. 3, pp. 50–57, 2017.
- [17] I. G. Cohen and M. M. Mello, “HIPAA and protecting health information in the 21st century,” *JAMA*, vol. 320, no. 3, pp. 231–232, 2018.
- [18] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
- [19] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” *Nature Medicine*, vol. 27, pp. 2176–2182, 2021.
- [20] M. Ghassemi, L. Oakden-Rayner, and A. L. Beam, “The false hope of current approaches to explainable AI in health care,” *The Lancet Digital Health*, vol. 3, no. 11, e745–e750, 2021.
- [21] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [22] World Health Organization, *Ethics and governance of artificial intelligence for health*, Geneva, 2021.
- [23] A. Jobin, M. Ienca, and E. Vayena, “The global landscape of AI ethics guidelines,” *Nature Machine Intelligence*, vol. 1, no. 9, pp. 389–399, 2019.
- [24] B. Mittelstadt, “Principles alone cannot guarantee ethical AI,” *Nature Machine Intelligence*, vol. 1, no. 11, pp. 501–507, 2019.
- [25] J. Morley, L. Floridi, L. Kinsey, and A. Elhalal, “From what to how: An initial review of publicly available AI ethics tools, methods and research,” *Science and Engineering Ethics*, vol. 26, pp. 2141–2168, 2020.
- [26] C. Petersen, J. Smith, R. R. Freimuth, et al., “Recommendations for the safe, effective use of adaptive CDS in the US healthcare system,” *JAMIA*, vol. 29, no. 4, pp. 677–684, 2022.
- [27] W. N. Price, S. Gerke, and I. G. Cohen, “Potential liability for physicians using artificial intelligence,” *JAMA*, vol. 322, no. 18, pp. 1765–1766, 2019.
- [28] OECD, *Oecd.ai policy observatory*, <https://oecd.ai/>.
- [29] National Institute of Standards and Technology, *Artificial intelligence risk management framework (AI RMF 1.0)*, 2023.